

PRAKTIKUM BASIS DATA

MODUL 9

SCRIPTING PROCEDURE AND FUNCTION



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Disusun Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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Diberikan Query sebagai berikut:

Query :

```
Worksheet: Query Builder

-- CREATE TABLE hremployees (
  employee_id      NUMBER(6)      NOT NULL,
  first_name       VARCHAR2(20),
  last_name        VARCHAR2(25)   NOT NULL,
  email            VARCHAR2(25)   NOT NULL,
  phone_number     VARCHAR2(20),
  hire_date        DATE           NOT NULL,
  job_id           VARCHAR2(10)   NOT NULL,
  salary           NUMBER(8,2),
  commission_pct   NUMBER(2,2),
  manager_id       NUMBER(6),
  department_id    NUMBER(4),
  CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
);

-- INSERT data untuk department_id = 90 (Executive -- dipakai soal no.4)
INSERT INTO hremployees VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', TO_DATE('17-JUN-2003', 'DD-MON-YYYY'), 'AD_PRES', 24000, NULL, NULL, 90);
INSERT INTO hremployees VALUES (101, 'Neena', 'Kochhar', 'NKOCHHAR', '515.123.4568', TO_DATE('21-SEP-2005', 'DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);
INSERT INTO hremployees VALUES (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', TO_DATE('13-JAN-2001', 'DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);

-- INSERT tambahan department lain (agar SELECT * tidak kosong)
INSERT INTO hremployees VALUES (103, 'Alexander', 'Hunold', 'AHUNOLD', '590.423.4567', TO_DATE('03-JAN-2006', 'DD-MON-YYYY'), 'IT_PROG', 9000, NULL, 102, 60);
INSERT INTO hremployees VALUES (104, 'Bruce', 'Ernst', 'BERNST', '590.423.4568', TO_DATE('21-MAY-2007', 'DD-MON-YYYY'), 'IT_PROG', 6000, NULL, 103, 60);
INSERT INTO hremployees VALUES (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2007', 'DD-MON-YYYY'), 'IT_PROG', 4200, NULL, 103, 60);
INSERT INTO hremployees VALUES (124, 'Trevin', 'Mourgos', 'TMOURGOS', '650.123.5234', TO_DATE('16-NOV-2007', 'DD-MON-YYYY'), 'ST_MAN', 5000, NULL, 100, 50);
INSERT INTO hremployees VALUES (141, 'Trenna', 'Rajs', 'TRAJS', '650.121.8009', TO_DATE('17-OCT-2003', 'DD-MON-YYYY'), 'ST_CLERK', 3500, NULL, 124, 50);
INSERT INTO hremployees VALUES (149, 'Ellen', 'Zlotkey', 'EZLOTKEY', TO_DATE('29-JAN-2008', 'DD-MON-YYYY'), 'SA_MAN', 10500, 2, 100, 80);
INSERT INTO hremployees VALUES (174, 'Ellen', 'Abel', 'EABEL', '011.44.1644.429267', TO_DATE('11-MAY-2004', 'DD-MON-YYYY'), 'SA_REP', 11000, 3, 149, 80);
```

Output :

[illegible]

Analisis :

Perintah tersebut digunakan untuk membuat tabel baru bernama employees dengan mengambil seluruh data dari tabel hrememployees.

Isi tabel hrememployees sendiri berisi data pegawai seperti:

- employee_id
- first_name
- last_name
- salary
- department_id
- dan lainnya.

Pada data yang diberikan terdapat:

- Department 90 → Executive
- Department 60 → IT
- Department 50 → Stock
- Department 80 → Sales

1. CREATE TABLE EMPLOYEES AS SELECT * FROM HREMPLOYEES

Query :

```
CREATE TABLE employees AS  
SELECT * FROM hremployees;
```

Output :

Table EMPLOYEES created.

Analisis:

Membuat tabel baru employees berdasarkan tabel hremployees.

- Seluruh data pada tabel sumber ikut dipindahkan
- Kolom pada tabel baru otomatis terbentuk mengikuti tabel asal
- Perubahan pada tabel employees tidak memengaruhi hremployees

2. SELECT * FROM EMPLOYEES

Query :

```
SELECT * FROM employees;
```

Output :

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100 Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101 Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102 Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90
4	103 Alexander	Hunold	AHUNOLD	590.423.4567	03-01-2006	IT_PROG	9000	(null)	102	60
5	104 Bruce	Ernst	BERNST	590.423.4568	21-05-2007	IT_PROG	6000	(null)	103	60
6	107 Diana	Lorentz	DLORENTZ	590.423.5567	07-02-2007	IT_PROG	4200	(null)	103	60
7	124 Kevin	Mourgos	KMOURGOS	650.123.5234	16-11-2007	ST_MAN	5800	(null)	100	50
8	141 Tenna	Rajs	TRAJS	650.121.8009	17-10-2003	ST_CLERK	3500	(null)	124	50
9	149 Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-01-2008	SA_MAN	10500	0,2	100	80
10	174 Ellen	Abel	EABEL	011.44.1644.429267	11-05-2004	SA_REP	11000	0,3	149	80

Analisis : Perintah ini digunakan untuk menampilkan seluruh isi tabel employees. Hasil yang muncul adalah seluruh data pegawai yang sebelumnya disalin dari tabel hremployees.

3. MEMBUAT PROCEDURE INCREASE_SALARY

Query :

```
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in IN NUMBER
)
IS
BEGIN
    FOR employee_rec IN (
        SELECT employee_id
        FROM employees
        WHERE department_id = increase_salary.department_id_in
    )
    LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
        WHERE emp.employee_id = employee_rec.employee_id;

        DBMS_OUTPUT.PUT_LINE('Updated ' || SQLROWCOUNT);
    END LOOP;
END increase_salary;
/
```

Output :

```
Procedure INCREASE_SALARY compiled
```

Analisis:

- Membuat procedure increase_salary untuk menaikkan salary pegawai
- Kenaikan salary berdasarkan department_id tertentu
- Menggunakan perulangan untuk update setiap pegawai
- Salary ditambah sesuai persentase kenaikan yang diinput
- Menampilkan jumlah data yang berhasil diupdate menggunakan DBMS_OUTPUT

4. MENJALANKAN PROCEDURE + ROLLBACK

Query :

```
BEGIN
    increase_salary (90, .50);
    ROLLBACK;
END;
/

SELECT * FROM employees
WHERE department_id = 90;
```

Output :

```
PL/SQL procedure successfully completed.
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

Analisis

Procedure dijalankan untuk:

- department 90
- kenaikan salary 50%

ROLLBACK digunakan untuk membatalkan seluruh perubahan transaksi.

Artinya:

- salary sempat berubah
- tetapi perubahan dibatalkan kembali
- data kembali ke salary awal

Sehingga hasil akhir SELECT tetap:

5. APA YANG TERJADI JIKA ROLLBACK DIHAPUS

Query :

```
ROLLBACK;  
  
SELECT * FROM employees  
WHERE department_id = 90;
```

Output :

SQL | All Rows Fetched: 3 in 0.037 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

Analisis:

- a. Perubahan salary tersimpan permanen
- b. Update data tidak dibatalkan
- c. Salary department 90 naik 50%
- d. Hasil SELECT menampilkan salary terbaru